

Stereo Distance Measurement with a Per-Camera Γ -Grid Model

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Abstract

We measure metric distance from two calibrated cameras with a small, fully transparent model. Each camera is a $5 \times 5 \times 2$ grid Γ of effective inverse focal lengths, interpolated bilinearly over the image, and the rig is a baseline $\mathbf{d}$ plus a relative rotation $R(\boldsymbol{\omega})$. We show that the Γ grid generalises radial lens distortion, and regularise it towards a learned radial field. The noise model (click and label noise) is estimated from the data and used for maximum-likelihood weighting, the regularisation strength is chosen by repeated cross-validation, and the robust objective is minimised by Levenberg–Marquardt with an analytic Jacobian. A Brown–Conrady model fitted to the same points is averaged in as a companion. All gradients are derived by hand; no automatic differentiation or SciPy is used. Under nested leave-one-out cross-validation on 27 real points, the error falls from 10.08 % (version 1) to **3.52 %** MAPE (median 1.96 %). This is close to the 3.13 % implied by the estimated measurement noise alone. On simulated rigs with exact ground truth, the method approaches the oracle error floor.

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1 Pixel and normalised coordinates

A pixel is (u, v) on a $W \times H$ sensor (2592×1944). With $(C_X, C_Y) = (W/2, H/2)$,

$$\tilde{u} = \frac{u - C_X}{C_X}, \quad \tilde{v} = \frac{v - C_Y}{C_Y} \quad \in [-1, 1]. \quad (1)$$

2 The ray equation

Every pixel defines a ray from the optical centre:

$$\mathbf{r}(u, v) = \begin{bmatrix} \tilde{u} \gamma_x(u, v) \\ \tilde{v} \gamma_y(u, v) \\ 1 \end{bmatrix}. \quad (2)$$

For an ideal pinhole with focal length f , $\gamma_x = C_X/f$ and $\gamma_y = C_Y/f$ everywhere. The nominal value is $\Gamma_0 = [C_X/f_0, C_Y/f_0]$ with $f_0 = 2880$ px. (Version 1 used one γ for both axes; since $C_X/C_Y = 4/3$ this scaled every vertical ray component by $4/3$.)

Proposition 1 (The Γ grid generalises radial distortion). *Let a lens map undistorted normalised coordinates $\mathbf{x}_u$ to distorted ones by $\mathbf{x}_d = \mathbf{x}_u D(\|\mathbf{x}_u\|)$ (radial distortion about the image centre, $D > 0$ and $\rho \mapsto \rho D(\rho)$ increasing). Then the exact ray of pixel (u, v) has the form (2) with*

$$\gamma_x(u, v) = \frac{C_X}{f} g(r_d), \quad \gamma_y(u, v) = \frac{C_Y}{f} g(r_d), \quad r_d = \frac{\sqrt{(u - C_X)^2 + (v - C_Y)^2}}{f},$$

for a single scalar function g of the distorted radius.

Proof. $\mathbf{x}_d = ((u - C_X)/f, (v - C_Y)/f)$ has norm $r_d = r_u D(r_u)$, which is invertible by monotonicity: $r_u = h(r_d)$. Then $\mathbf{x}_u = \mathbf{x}_d / D(h(r_d))$. With $g := 1/(D \circ h)$ we get $x_u = \tilde{u} (C_X/f) g(r_d)$ and likewise for y . $\square$

So any radial lens is a *radially symmetric* Γ field; tangential distortion and manufacturing imperfections make it anisotropic and non-radial, which a grid can still follow. For $D = 1 + k_1 r^2 + k_2 r^4$, $g(r) = 1 - k_1 r^2 + (3k_1^2 - k_2) r^4 + O(r^6)$: this is the form of the radial prior of §6.1.

3 Bilinear interpolation

Grid nodes sit at $u = j c_w$, $v = i c_h$ with $c_w = W/4$, $c_h = H/4$. With $f_x = u/c_w$, $f_y = v/c_h$, cell $(i_y, i_x) = (\lfloor f_y \rfloor, \lfloor f_x \rfloor)$ (clamped) and $t_x = f_x - i_x$, $t_y = f_y - i_y$:

$$\gamma(u, v) = \sum_{k=1}^4 w_k \Gamma_{n_k}, \quad \mathbf{w} = ((1 - t_x)(1 - t_y), t_x(1 - t_y), (1 - t_x)t_y, t_x t_y). \quad (3)$$

The weights sum to one, reproduce fields linear in (u, v) exactly, make γ continuous, and give $\partial\gamma/\partial\Gamma_{n_k} = w_k$. Noise-free experiments (§14) show that the 5×5 grid represents a distorted lens to 0.06% depth error, so the representation is not the limiting factor.

4 Stereo geometry

4.1 Rig model and ray-space rectification

For one 3-D point, $P_L = R(\omega)P_R + \mathbf{d}$ with $\mathbf{d} = [d_x, d_y, 0]^\top$ and $R = R_z(\omega_{\text{roll}})R_y(\omega_{\text{yaw}})R_x(\omega_{\text{pitch}})$. The rig has about 1° of yaw (a constant 61 px disparity offset) and about 1° of roll (vertical

disparity linear in u). A multiplicative γ can produce neither. The right ray is rotated and perspective-normalised,

$$\mathbf{w} = R\mathbf{r}_R, \quad \hat{\mathbf{r}}_R = \mathbf{w}/w_z, \quad (4)$$

so that $Z\mathbf{r}_L - Z'\hat{\mathbf{r}}_R = \mathbf{d}$ and, because $d_z = 0$, $Z = Z'$:

$$r_{L,x} - \hat{r}_{R,x} = \frac{d_x}{Z}, \quad r_{L,y} - \hat{r}_{R,y} = \frac{d_y}{Z}. \quad (5)$$

The derivatives $\partial R/\partial \omega_j$ are products of the elementary rotations and one derivative factor. They are exact, and checked against finite differences.

4.2 Triangulation

$A\mathbf{x} = \mathbf{d}$ with $A = [\mathbf{r}_L \quad -\hat{\mathbf{r}}_R]$, $\mathbf{x} = [Z, Z']^\top$ is solved by the normal equations $A^\top A\mathbf{x} = A^\top \mathbf{d}$ (closed-form 2×2 inverse, vectorised). The same problem is also solved by batch gradient descent on $\|A\mathbf{x} - \mathbf{d}\|^2$ with step $1/L$, $L = 2\lambda_{\max}(A^\top A)$. Plain GD contracts by $1 - 1/\kappa$ per step, and for distant points κ reaches 10^3 – 10^4 , so Nesterov momentum $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ is used ($O(\sqrt{\kappa})$ steps). The two agree to 10^{-8} (tested), which is why OLS is the authoritative path. Since $Z = d_x/D$, $\partial Z/\partial D = -Z^2/d_x$: depth error grows with Z^2 .

5 Loss function and noise model

5.1 Residuals

$$e_x = D_x \frac{Z}{d_x} - 1, \quad e_y = D_y \frac{Z}{d_x} - \frac{d_y}{d_x}, \quad D_{x,y} = r_{L,x/y} - \hat{r}_{R,x/y}. \quad (6)$$

e_x is the relative disparity error ($\approx$ relative depth error). It avoids the Z^2 gradient blow-up of a $(Z_{\text{pred}} - Z)^2$ loss. e_y is the epipolar misalignment in the same units.

5.2 Noise model and maximum-likelihood weights

Two independent sources perturb a calibration point: click noise (σ_{px} per coordinate), which changes the ray-space disparity by $\sqrt{2}\sigma_{\text{px}}/f$; and label noise (the tape measurement), a relative error σ_ℓ . Propagating both through (6),

$$\text{Var}(e_x) \approx \sigma_\ell^2 + c^2 Z^2, \quad \text{Var}(e_y) \approx c^2 Z^2, \quad c = \frac{\sqrt{2}\sigma_{\text{px}}}{f d_x}. \quad (7)$$

Equal weighting of e_x (version 2.0) is therefore maximum-likelihood only if label noise dominates. Following feasible generalised least squares, we (i) fit with equal weights, (ii) estimate c robustly from e_y and σ_ℓ from the remainder of e_x (median-absolute-deviation estimators, robust to mis-clicks), and (iii) refit with residuals scaled by $1/\sigma_i$:

$$\tilde{e}_{x,i} = e_{x,i} \sqrt{\frac{\kappa^2 + Z_{\text{ref}}^2}{\kappa^2 + Z_i^2}}, \quad \tilde{e}_{y,i} = e_{y,i} \frac{Z_{\text{ref}}}{Z_i}, \quad \kappa = \sigma_\ell/c, \quad (8)$$

normalised at $Z_{\text{ref}} = \text{RMS}(Z)$ so the Huber threshold keeps its meaning. On the real data the estimate is $\sigma_\ell = 1.0\%$, $\sigma_{\text{px}} = 5.3 \text{ px}$, $\kappa = 0.86 \text{ m}$. On synthetic data with known noise, the estimator recovers both within 35% (tested).

5.3 Robust objective

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left[\rho_\delta(\tilde{e}_{x,i}) + \lambda_y \rho_\delta(\tilde{e}_{y,i}) \right] + \frac{1}{2}(\theta - \theta_0)^\top H_{\text{reg}}(\theta - \theta_0), \quad (9)$$

with the Huber function ρ_δ ($\delta = 0.05$), $\lambda_y = 0.5$, and the regulariser of §6.

6 Priors, gauge and initialisation

6.1 Hierarchical radial prior

Version 2.0 shrank Γ towards the constant Γ_0 (anchor λ_a) and penalised its membrane energy $\frac{1}{2} \sum_{\text{edges}} (\Gamma_i - \Gamma_j)^2 = \frac{1}{2} \Gamma^\top \mathbf{L} \Gamma$ ($\mathbf{L}$: graph Laplacian). Every real lens has a radial "bowl" (Prop. 1), which this prior penalises, and that biases the fit when data are few. Version 2.1 shrinks each camera's grid towards a *learned* radial field:

$$\Gamma_{\text{prior}}(n) = \Gamma_0(1 + a_1 r_n^2 + a_2 r_n^4), \quad \mathcal{R} = \frac{1}{2} \boldsymbol{\delta}^\top (\lambda_a I + \lambda_s \mathbf{L}) \boldsymbol{\delta}, \quad \boldsymbol{\delta} = \Gamma - \Gamma_0 - B \mathbf{a}, \quad (10)$$

where r_n is the node's distorted radius and B stacks $\Gamma_0[r_n^2, r_n^4]$. $\mathcal{R}$ is an exact quadratic form in $(\Gamma, \mathbf{a})$ with constant Hessian $M^\top (\lambda_a I + \lambda_s \mathbf{L}) M$, $M = [I, -B]$. The coefficients $\mathbf{a}$ enter only the prior, never the rays, so inference and the file format are unchanged. With little data Γ follows the two-parameter radial field; with more data it departs from it wherever the data demand. This is the classical parametric model used as a prior mean for a non-parametric one.

6.2 Scale gauge and its sensitivity

Scaling every γ and d_x by one factor leaves (5) unchanged for a parallel rig, so the centre node of Γ_L is frozen at Γ_0 (nominal focal length f_0). With a rotation the invariance is only approximate (the perspective division mixes the scaled and unscaled components), and the priors are expressed in Γ units, so f_0 is weakly identifiable rather than a pure gauge. We measured the consequence: on the real data, refitting with $f_0 \in \{2500, 2880, 3300\}$ px shifts individual depths by at most 0.8%; on a simulated rig with true focal lengths 2750/2790 px, choosing f_0 anywhere in $[2200, 3800]$ px changes the held-out MAPE by at most 0.04 percentage points. Accuracy is therefore insensitive to f_0 ; d_x is the physical baseline only if the true focal length equals f_0 . On a distortion-free rig with f_0 correct, training recovers d_x , d_y and all three angles exactly (tested).

6.3 Closed-form warm start

Linearising $R \approx I + [\boldsymbol{\omega}]_\times$ with nominal Γ gives two equations per point that are linear in $(d_x, d_y, \omega_p, \omega_y, \omega_r)$,

$$r_{L,x} - r_{R,x} = d_x/Z + \omega_y - \omega_r r_{R,y}, \quad r_{L,y} - r_{R,y} = d_y/Z - \omega_p + \omega_r r_{R,x},$$

solved by weighted least squares (DLT-style initialisation).

7 Analytic gradients and Jacobian

With $s = Z/d_x$ and $\mathbf{a} = \partial \mathcal{L} / \partial \hat{\mathbf{r}}_{R,xy}$:

$$\frac{\partial e_x}{\partial \Gamma_{L,n_k,x}} = s \tilde{u}_L w_k, \quad \frac{\partial e_y}{\partial \Gamma_{L,n_k,y}} = s \tilde{v}_L w_k, \quad (11)$$

$$\frac{\partial \hat{r}_x}{\partial \mathbf{w}} = \left[\frac{1}{w_z}, 0, -\frac{w_x}{w_z^2} \right], \quad \frac{\partial \hat{r}_y}{\partial \mathbf{w}} = \left[0, \frac{1}{w_z}, -\frac{w_y}{w_z^2} \right], \quad \frac{\partial \hat{r}}{\partial \mathbf{r}_R} = \frac{\partial \hat{r}}{\partial \mathbf{w}} R, \quad \frac{\partial \hat{r}}{\partial \omega_j} = \frac{\partial \hat{r}}{\partial \mathbf{w}} \frac{\partial R}{\partial \omega_j} \mathbf{r}_R, \quad (12)$$

$$\frac{\partial e_x}{\partial d_x} = -\frac{e_x + 1}{d_x}, \quad \frac{\partial e_y}{\partial d_x} = -\frac{e_y}{d_x}, \quad \frac{\partial e_y}{\partial d_y} = -\frac{1}{d_x}, \quad (13)$$

and the right grid receives $-s (\partial \hat{r} / \partial \mathbf{r}_R)_x \tilde{u}_R w_k$ (and the y analogue). These give both the gradient of $\mathcal{L}$ (for Adam) and the residual Jacobian J (for Levenberg–Marquardt). All components agree with central finite differences (largest relative discrepancy 10^{-7} for the gradient, $5 \cdot 10^{-6}$ for J).

8 Optimisation

8.1 Levenberg–Marquardt with IRLS (default)

Multiplying (9) by N , each iteration solves

$$(J^\top W J + N H_{\text{reg}} + \mu \text{diag}) \Delta = -(J^\top W \tilde{\mathbf{e}} + N H_{\text{reg}}(\theta - \theta_0)), \quad (14)$$

where W holds the Huber IRLS weights $\min(1, \delta/|\tilde{e}|)$ and λ_y . A step is accepted only if the objective decreases ($\mu \leftarrow 0.3\mu$), otherwise $\mu \leftarrow 5\mu$, so the method is monotone. A fit on 27 points takes about 0.04 s.

8.2 Adam

Adam with bias correction, per-block learning rates and cosine decay minimises the same objective. On the identical objective, both optimisers reach the same minimum to 10^{-10} relative (Fig. 1, and tested), and the gradient vanishes at the LM solution ($< 10^{-7}$). Two independent methods agreeing certifies convergence.

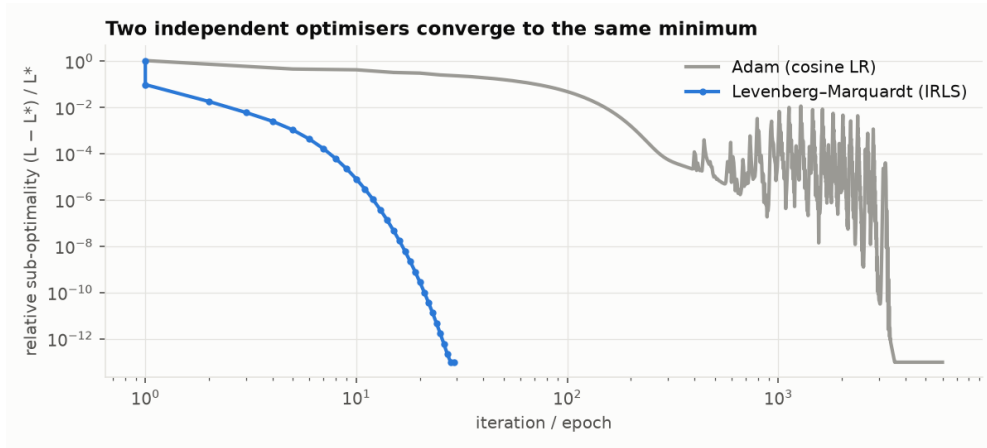


Figure 1: Relative sub-optimality of LM and Adam on the same objective.

9 Model selection and model averaging

9.1 Regularisation strength by repeated cross-validation

Noise-free experiments show that the best λ_s depends strongly on N and the noise level, so it cannot be fixed. `train` selects $\lambda_s \in \{0.01, \dots, 3\}$ by 5-fold cross-validation repeated over 3 shuffles (held-out depth MAPE; ties go to the stronger penalty), using *only its training points*. Evaluating this procedure by leave-one-out is therefore a *nested* cross-validation (Varma & Simon 2006): no hyper-parameter ever sees the test point. On all 27 points it selects $\lambda_s = 1$.

9.2 Averaging with a Brown–Conrady companion

The standard parametric alternative is the Brown–Conrady model used by OpenCV. Per camera it has a focal length, a principal point, radial k_1, k_2 and tangential p_1, p_2 , plus the same rotation and baseline (18 parameters, $f_L = f_0$ as gauge). It is fitted with *the same residuals, Huber loss and noise model*. Its out-of-sample errors correlate with the Γ grid’s at only $\rho = 0.55$. For two unbiased predictors of equal error variance σ^2 , the equal-weight average has variance $\frac{\sigma^2}{2}(1 + \rho)$ (Bates & Granger 1969). Averaging therefore reduces error unless $\rho \approx 1$. `v2.1` reports the equal-weight average of both depths. The weights are fixed in advance, not fitted, so the averaging

introduces no selection bias. The companion is used only from 25 calibration points: with fewer, its 18 parameters are poorly determined and averaging hurts (synthetic study, §14). On simulated lenses the Γ grid alone is often the best single model (it beats Brown–Conrady in 16 of 20 settings); on the real rig the average is better, which we attribute to real-world effects outside either model. The Γ grid remains the geometric engine (epipolar curves, matching, uncertainty).

10 Diagnostics

Effective degrees of freedom. For the penalised fit, $\text{df} = \text{tr}[(J^\top W J + N H_{\text{reg}})^{-1} J^\top W J]$ (the trace of the hat matrix). The model has 109 parameters (107 free), but $\text{df} = \mathbf{15.1}$: the data determine about fifteen directions in parameter space, and the priors fix the rest.

Parameter uncertainty. The sandwich covariance $A^{-1}(\sigma_x^2 J_x^\top W_x J_x + \sigma_y^2 J_y^\top W_y J_y) A^{-1}$ is compared with a 200-sample nonparametric bootstrap:

Rig parameter	value	s.e. (sandwich)	s.e. (bootstrap)
dx (m)	0.2148	0.0060	0.0086
dy (m)	0.0014	0.0020	0.0118
pitch (°)	0.254	0.071	0.242
yaw (°)	1.099	0.096	0.181
roll (°)	-1.196	0.186	0.236

The two agree for d_x , yaw and roll. For d_y and pitch, which are identified only by the heavy-tailed vertical residuals, the sandwich estimate is optimistic; the bootstrap values should be quoted.

11 Epipolar geometry and correspondence search

For a left pixel and depth Z : $P_R = R^\top(Z\mathbf{r}_L - \mathbf{d})$, then Γ_R is inverted by the fixed point $u \leftarrow C_X + C_X t_x / \gamma_x(u, v)$. Sweeping $1/Z$ traces the curved, tilted epipolar curve. Along it, zero-mean normalised cross-correlation (31×31 patches, every pixel) is maximised and refined with the parabola $\Delta = (s_{-1} - s_{+1}) / (2(s_{-1} - 2s_0 + s_{+1}))$. A match is accepted only if its ZNCC is ≥ 0.6 , the template is textured, and the reverse search along the *left* epipolar curve returns within 2 px of the click (a left–right consistency check). Accepted matches are refined by inverse-compositional Lucas–Kanade. Images are pre-processed by CLAHE with bilinear blending between tiles.

12 Uncertainty of a measurement

Click noise is propagated to first order through the complete model, and the model’s relative prediction error ε (from the cross-validation inside training) is added in quadrature:

$$\sigma_Z^2 = \sum_{p \in \{u_L, v_L, u_R, v_R\}} \left(\frac{\partial Z}{\partial p} \right)^2 \sigma_{\text{px}}^2 + (\varepsilon Z)^2.$$

Out of sample, $|Z - Z_{\text{true}}| \leq \sigma_Z$ holds for 85 % of points (Gaussian ideal 68 %) and $\leq 2\sigma_Z$ for 96 % (ideal 95 %), with RMS z -score 0.80. The interval is conservative at 1σ because the errors are heavy-tailed, and well calibrated at 2σ .

13 What is measured, and verification

Quantities. The *depth* Z is the distance along the left optical axis; the calibration labels are depths (they fit better than straight-line ranges, 5.1 % vs 5.9 % LOOCV). The 3-D point is $P =$

$Z \mathbf{r}_L = [X, Y, Z]^\top$ and the straight-line distance from the left optical centre is $\|P\| = Z\|\mathbf{r}_L\|$, which exceeds Z by up to 21 % in the image corners; the app reports both, and the length $\|P_1 - P_2\|$ between two measured points with $\sigma_\ell^2 = \sum_k (\sigma_{Z_k} \mathbf{u}^\top P_k / Z_k)^2$ ($\mathbf{u}$ the unit segment direction). Depths outside $[0.8 Z_{\min}, 1.25 Z_{\max}]$ of the calibration are flagged as extrapolated.

Lateral scale. Depth labels identify only the product $f d_x$: scaling every γ by α and d_x by α leaves all depths (and hence the whole calibration objective, up to the weak rotation coupling) unchanged, but multiplies $X = Z \tilde{u} \gamma_x$ and Y by α . The lateral coordinates are therefore fixed by the assumed f_0 and are off by f_{true}/f_0 . We restore them with a lateral scale k ($X, Y \mapsto kX, kY$) fitted from reference lengths L_j between measured point pairs, $\min_k \sum_j (\|(k\Delta X_j, k\Delta Y_j, \Delta Z_j)\| - L_j)^2$ (Gauss–Newton), or by supplying the true focal length. On a simulated rig with $f_{\text{true}} = 2750$ px and $f_0 = 2880$ px, lengths are 2 % short before and within 0.4 % after three references; depths are unchanged (tested).

Verification. `scripts/audit.py` and the test-suite check that (i) projecting a 3-D point into the right image and triangulating it back reproduces the depth to $7 \cdot 10^{-9}$ (shipped calibration, 1953 random points, 0.3–30 m); (ii) on a distortion-free simulated rig with known geometry, training recovers $d_x = 0.25000$ m, d_y and all three angles exactly; (iii) the full pipeline on a ray-traced scene (click $\rightarrow$ ZNCC $\rightarrow$ left–right check $\rightarrow$ Lucas–Kanade $\rightarrow$ depth) is within 3 % of the true depth for 96.1 % of accepted matches, and straight-line distances have a median error of 0.5 %; (iv) gradients, Jacobian, the two optimisers and the two triangulation solvers agree as stated above; (v) no library vision, calibration or machine-learning algorithm is used: all of the geometry, optimisation, matching and statistics is implemented in this repository on top of NumPy linear algebra (OpenCV only reads images in the labelling tool, Pygame only draws the interface).

14 Evaluation

14.1 Protocol

With $N = 27$ labelled points all real-data numbers are nested leave-one-out cross-validation. Confidence intervals come from 4000 bootstrap resamples. Paired comparisons use a two-sided sign-flip permutation test (20 000 draws, no normality assumption) with Holm’s correction for the five comparisons.

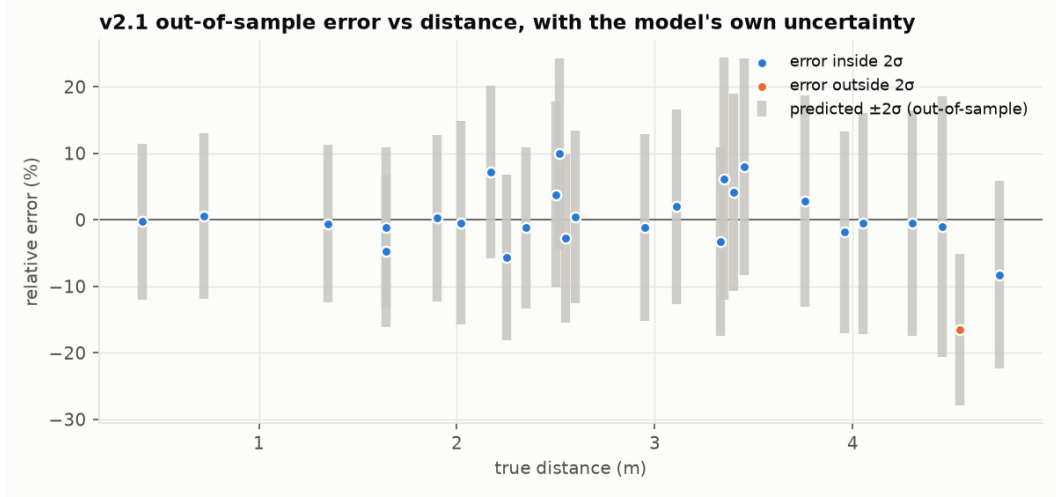
14.2 Real dataset

Model	MAPE	95 % CI	median	$\delta < 1.05$	worst
v1 original (reproduced exactly)	10.08 %	5.7–16.3	5.98 %	37 %	77 %
Pinhole + disparity offset (2 params)	5.90 %	4.3–7.7	5.09 %	48 %	16 %
Brown–Conrady, 18 params (same loss, FGLS)	4.20 %	2.8–5.8	3.04 %	78 %	14 %
Γ grid + rotation, v2.0 loss	5.64 %	3.7–7.8	4.13 %	56 %	21 %
+ estimated noise model (FGLS)	5.27 %	3.4–7.4	3.91 %	63 %	22 %
+ radial prior (Γ grid alone)	4.58 %	3.3–6.1	3.23 %	67 %	19 %
+ Brown–Conrady companion (v2.1)	3.52 %	2.2–5.0	1.96 %	74 %	17 %
v2.1 without rig rotation	4.86 %	3.5–6.5	3.22 %	59 %	17 %

Comparison	Δ MAPE (pp)	p	p_{Holm}
v2.1 vs v1	−6.56	0.0007	0.0035
v2.1 vs Brown–Conrady	−0.68	0.1676	0.4860
v2.1 vs Γ alone	−1.06	0.0243	0.0970
Γ alone vs Brown–Conrady	+0.38	0.6374	0.6374
Γ + FGLS vs Γ v2.0 loss	−0.37	0.1620	0.4860

v2.1 is significantly better than version 1. Among the modern models the differences are not significant with 27 points; they are resolved by the controlled benchmark below. Removing the rig rotation costs over one percentage point.

Error budget. The noise model estimated from the data predicts 2.99 % MAPE from click noise alone and 3.13 % with label noise, against the observed 3.52 %. Most of the remaining error is therefore measurement noise in the calibration data, not model error. Further gains on this rig require better data (Section 16).



14.3 Controlled synthetic benchmark

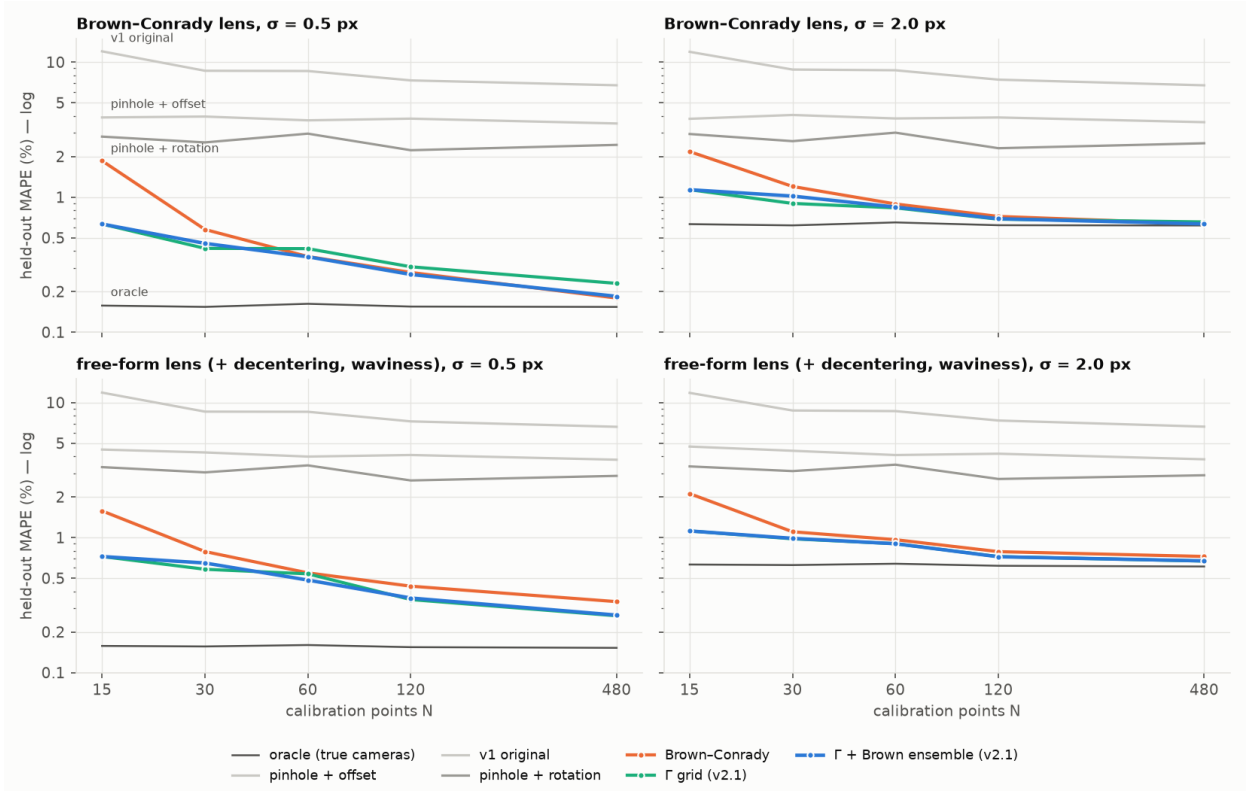
Two simulated rigs have exact ground truth: a Brown–Conrady lens (the baseline’s own model family), and a free-form lens that adds decentering and a smooth non-radial waviness of ± 4 px. Both have unequal focal lengths, off-centre principal points, a 0.20 m baseline and a $(0.15^\circ, 0.9^\circ, -1.1^\circ)$ rotation. Training points carry click noise σ and 1 % label noise; 500 fresh points per seed are scored. The *oracle* triangulates with the true cameras, so its error is the floor set by the test clicks’ own noise. Held-out MAPE (%), best fitted model in bold:

<i>Brown–Conrady lens</i>					
$\sigma = 0.5$ px, $N =$	15	30	60	120	480
oracle (true cameras)	0.16	0.15	0.16	0.15	0.15
pinhole + offset	3.90	3.95	3.71	3.82	3.51
v1 original	12.02	8.63	8.60	7.32	6.74
pinhole + rotation	2.81	2.54	2.95	2.23	2.44
Brown–Conrady	1.86	0.58	0.36	0.28	0.18
Γ grid alone	0.63	0.42	0.42	0.31	0.23
v2.1 ensemble	0.63	0.45	0.36	0.27	0.18
$\sigma = 2.0$ px, $N =$	15	30	60	120	480
oracle (true cameras)	0.63	0.62	0.65	0.62	0.62
pinhole + offset	3.80	4.06	3.83	3.89	3.59
v1 original	11.92	8.81	8.70	7.42	6.73
pinhole + rotation	2.94	2.60	3.00	2.30	2.51
Brown–Conrady	2.18	1.20	0.89	0.72	0.63
Γ grid alone	1.14	0.90	0.83	0.69	0.66
v2.1 ensemble	1.14	1.02	0.84	0.69	0.63

Free-form lens

$\sigma = 0.5 \text{ px}, N =$	15	30	60	120	480
oracle (true cameras)	0.16	0.16	0.16	0.15	0.15
pinhole + offset	4.49	4.27	3.98	4.09	3.77
v1 original	11.85	8.57	8.55	7.25	6.61
pinhole + rotation	3.32	3.04	3.43	2.65	2.87
Brown–Conrady	1.58	0.79	0.55	0.44	0.34
Γ grid alone	0.72	0.58	0.54	0.35	0.26
v2.1 ensemble	0.72	0.65	0.48	0.36	0.27

$\sigma = 2.0 \text{ px}, N =$	15	30	60	120	480
oracle (true cameras)	0.63	0.63	0.64	0.62	0.61
pinhole + offset	4.72	4.39	4.09	4.18	3.80
v1 original	11.80	8.73	8.64	7.36	6.63
pinhole + rotation	3.37	3.11	3.47	2.72	2.89
Brown–Conrady	2.12	1.10	0.96	0.79	0.72
Γ grid alone	1.12	0.98	0.90	0.72	0.67
v2.1 ensemble	1.12	0.99	0.90	0.72	0.67



14.4 Automatic matching

On a ray-traced scene seen through the simulated rig, 200 random clicks: 77% are accepted by ZNCC + left-right check; 96.1% of those are within 3% of the true depth (median error 0.41%); 15 ms per click.

15 Complexity and numerical safeguards

Ray evaluation and triangulation are $O(1)$. One LM iteration costs $O(NP^2 + P^3)$ with $P = 109$ (well under a millisecond). A full training run (noise model, 90 inner fits for λ_s , companion) takes about 3s, and nested LOOCV about 2min. Safeguards: near-singular $A^\top A$ falls back to

$Z = d_x/D$; $D \leq 0$ or $Z \leq 0$ gives NaN; $\Gamma \in [0.05, 5]$ and $d_x \geq 0.02$ m; LM only accepts decreasing steps; trial steps that make the Brown–Conrady undistortion diverge are rejected.

16 Limitations and future work

- **Data.** The error budget shows that the 27 hand-clicked points, with estimated $\sigma_{\text{px}} \approx 5.3$ px and $\sigma_\ell \approx 1.0\%$, limit accuracy. A calibration board at measured distances would give hundreds of sub-pixel points; the synthetic results suggest errors near 0.3% would then be reachable.
- Model differences below about one percentage point cannot be resolved with 27 points (the CIs are about ± 1.5 pp).
- Bundle adjustment (using correspondences without known distance) and multi-scale grids are natural extensions.

A Mapping between code and mathematics

Function (package <code>stereo_gamma</code>)	Implements
<code>geometry.bilinear_weights</code> , <code>rays</code>	eqs. (3), (2)
<code>geometry.rotation_with_jacobians</code> , <code>right_rays_in_left</code>	R , $\partial R/\partial \omega$, eq. (4)
<code>model.StereoModel.triangulate_gamma</code> , <code>triangulate_gd</code>	OLS; batch GD + Nesterov
<code>model.StereoModel.triangulate</code>	v2.1 ensemble depth
<code>model.StereoModel.depth_uncertainty</code>	error propagation
<code>training.objective</code> , <code>solver.residual_jacobian</code>	eq. (9), gradient and J
<code>training.estimate_noise</code> , <code>noise_scales</code>	noise model, eq. (7)
<code>solver.regulariser_hessian</code> , <code>radial_basis</code>	anchor, membrane, radial prior
<code>solver.levenberg_marquardt</code> , <code>training.Adam</code>	optimisers
<code>training.select_smoothing</code>	repeated k -fold selection of λ_s
<code>solver.diagnostics</code>	effective df, sandwich covariance
<code>baselines.BrownConradyStereo</code>	parametric companion / baseline
<code>matching.epipolar_match</code> , <code>lk_refine</code> , <code>clahe</code>	correspondence search
<code>evaluation.loocv</code> , <code>coverage</code> , <code>signflip_test</code>	evaluation protocol
<code>legacy.train_legacy</code>	exact re-implementation of version 1
<code>synthetic.SyntheticRig.triangulate</code>	oracle

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